

Ali Wael Mohamed Ali Ahmed

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Professional Summary

Passionate Mechatronics Engineer with a robust background bridging **Embedded Systems, Robotics, Industrial Automation, and Control Theory**. Experienced across the complete multidisciplinary product development cycle, from electronic hardware design and PCB integration to writing robust C/C++ firmware, implementing closed-loop control algorithms, and deploying Embedded Linux applications. Proven track record in microcontroller architecture, communication protocols (CAN, SPI, I2C), PLC programming (TIA Portal), and robot kinematics. Adept at translating complex physical system constraints into reliable, high-performance software and automation solutions.

Education

Cairo University, Giza, Egypt

B.Sc. in Mechatronics Engineering — Graduated: July 2026

Relevant Coursework: Control Systems, Automatic Control, Advanced Robotics, Embedded Systems, Microprocessors & Microcontrollers, Design of Mechatronics Systems, Pattern Recognition & Machine Learning, Electric Drive.

Professional Experience

Embedded Linux Engineer Intern | Ezzmedical, Cairo, Egypt *(Aug 2024, Aug 2025 – Sep 2025)*

- Developed and deployed a custom Linux OS for EZvent (Egypt's first unconditionally approved medical ventilator), building tailored Yocto images and toolchains for Raspberry Pi 4 and iMX8mp boards.
- Executed hardware-software integration, deploying a Qt-based GUI directly onto embedded ventilator boards.
- Optimized low-level system performance and resolved critical build challenges to ensure system stability for life-critical operations.

Level 2 Automation Engineer Intern | Al Ezz Dekheila Steel Co. (EZDK), Alexandria, Egypt *(Jul 2024)*

- Mapped industrial data flows between Level 1 (PLC/Field instrumentation) and Level 2 automation hierarchies.
- Documented real-time data acquisition strategies across distributed industrial systems, gaining practical insight into industrial communication and plant-wide monitoring.

Technical Projects

Machine Learning Enhanced Emergency BVM Ventilator (Team Lead) *(Jan 2026 – May 2026)*

- **Hardware & PCB Design:** Designed the full electrical schematic and contributed to the PCB layout. Engineered the power management system with custom crowbar protection circuits to safely deliver power to a 36V stepper motor, Raspberry Pi, and Arduino Uno simultaneously.
- **Embedded Firmware & Sensors:** Developed safety-critical C++ firmware utilizing a 3-layer architecture. Implemented signal conditioning and EMA filtering for I2C environmental (BMP280) and differential pressure flow sensors.
- **Control & Motion Profiling:** Implemented S-Curve motion profiling for stepper motor actuation to guarantee smooth, jerk-free volume delivery under dynamic patient lung resistance.
- **Linux & GUI Integration:** Built a Python/Qt GUI application on a touchscreen display (Wayland/X11), configuring systemd auto-boot services and low-latency serial communication.
- **Full-Cycle Leadership:** Led an interdisciplinary engineering team through the complete product development lifecycle, from component selection and hardware validation to electromechanical testing.

Pick and Place Robotic Arm & Custom Gripper Control *(Apr 2026 – May 2026)*

- **Custom Gripper & Closed-Loop PID Control:** Engineered a custom robotic gripper mechanism driven by an open-loop DC motor. Implemented a closed-loop PID control system utilizing potentiometer feedback for precise position regulation and motor control.
- **Motion Profiling & Kinematics:** Developed synchronized multi-axis interpolation algorithms for the robot's servo motors, ensuring fast, smooth, and vibration-free movement profiles.
- **Computer Vision Integration:** Integrated an OpenCV face mask detection model to demonstrate automated visual interaction and conditional pick-and-place execution.

Robotic Arm Trajectory Automation & Teleoperation *(Nov 2025 – Dec 2025)*

- **CAN Bus Integration:** Configured low-level CAN bus network interfaces within a Linux environment to handle high-speed live telemetry and real-time state feedback for a 6-DOF robotic manipulator.
- **Control Laws & Teleoperation:** Translated physical joint mechanics into software control laws using the robot's "teach mode" for high-precision trajectory tracking; integrated HTTP requests for smartphone IMU-based remote teleoperation.

Autonomous Museum Guide Robot & AI Integration *(Feb 2025 – May 2025)*

- **Control & Sensor Fusion:** Interfaced an MPU6050 IMU and rotary encoders to the main control unit, applying PID control loops for stable real-time autonomous navigation, odometry, and Go-To-Goal logic.
- **Mechanical Design & Hardware Integration:** Designed the robot's mobile

chassis in SOLIDWORKS for laser cutting, mounting custom PCBs, motor drivers, and power distribution electronics.

- **Machine Learning & Final Delivery:** Deployed a custom-trained YOLOv11 object detection model for real-time artifact recognition, integrating a Text-to-Speech (TTS) pipeline for interactive tourist guidance.

Automated Bottling & Capping Line Simulation *(Dec 2024 – Jan 2025)*

- **PLC Programming & Automation Logic:** Programmed Siemens PLCs (Ladder Logic) in TIA Portal to simulate full sequential bottling, filling, and capping operations with safety interlocks and reset routines.
- **HMI & Process Monitoring:** Designed an interactive HMI interface for real-time process monitoring, manual override simulations, and fault alarms.

Minimalist Smartwatch on STM32 Microcontroller *(Jun 2024)*

- **Firmware & Microcontroller Architecture:** Built a real-time digital watch utilizing bare-metal ARM Assembly programming on an STM32F103C8T6 microcontroller.
- **Hardware Fundamentals:** Leveraged direct register manipulation, memory mapping, interrupt service routines (ISRs), and hardware timers to achieve sub-1-second/day accuracy; interfaced a TFT display via optimized GPIO protocols.

Machine Fault Recognition using CNN *(Mar 2026 – May 2026)*

- **Deep Learning for Diagnostics:** Developed an end-to-end Convolutional Neural Network (CNN) model for machine vibration/fault recognition.
- **ML Pipeline Implementation:** Executed comprehensive data preprocessing, feature extraction, model training, and inference testing applicable to industrial predictive maintenance and condition monitoring.

Core Skills

- **Embedded Systems & Firmware:** C/C++, Bare-Metal ARM Assembly, Embedded Linux (Yocto, Buildroot), Hardware Abstraction Layers (HAL), Microcontroller Architecture (STM32, ESP32, Arduino, Raspberry Pi).
- **Robotics & Control Theory:** Classical & Modern Control, Closed-Loop PID Control, Motion Profiling (S-Curve, Multi-Axis Interpolation), Forward & Inverse Kinematics, Sensor Fusion (IMU, Encoders), ROS.
- **Industrial Automation:** PLC Programming (Siemens TIA Portal, Ladder Logic), HMI Design, Level 1 & Level 2 Industrial Automation Hierarchies, Industrial Sensors & Actuators.
- **Electronics & Hardware Design:** Schematic Design & PCB Layout basics (KiCad), Circuit Protection (Crowbar Circuits), Power Management, Sensor Interfacing, Hardware-Software Integration & Debugging.
- **Communication Protocols:** CAN Bus, I2C, SPI, UART, PWM, ADC, HTTP/IoT.
- **Software & Tools:** Linux/Bash, Git/GitHub, SOLIDWORKS, MATLAB / Simulink, Keil uVision, TIA Portal, OpenCV, Python.

Certifications & Languages

- **Certifications:** Embedded Linux Course (IMT, 2023) | DELF B2 (French Ministry of Education, 2021)
- **Languages:** Arabic (Native), English (Proficient - Spoken & Written), French (Advanced/B2), Spanish (Beginner)